UNIVERSITY OF COMPUTER STUDIES, YANGON

Cross-Disciplinary and Collaborative Project

Software Design Model for

Treasurer's Hub
Online Shopping System

Covered By

Design and Architecture with UML

Faculty of Information Science

Submitted By

Group No - 2

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ABSTRACT

This project is the increment of the requirement analysis model for Treasurer's Hub Online Shopping Model which is constructed and submitted in the partial completion of Data Modeling in Software Engineering course. In this project, the former analysis model is re-evaluated through a series of discussion in the team and is refined to include software design aspects. The team would like to express special thanks to our teacher Prof. Dr. Zin Thu Thu Myint from the Faculty of Information Science for giving us guidelines and supervision.

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Chapter 1
Introduction

In today's digital age, online shopping systems have become a ubiquitous part of our lives. To effectively model these complex platforms, we need a comprehensive approach that captures the system's functionality, data structures, and user interactions. This introduction delves into the various models used to represent online shopping systems, exploring how each model provides a unique perspective on the system's inner workings.

We'll examine use cases that define user journeys, class diagrams that visualize system components, sequence diagrams that detail message flows, and ER diagrams that map out data relationships. By understanding these models, we can design, develop, and maintain online shopping systems that are not only user-friendly but also efficient, secure, and scalable.

1.1 Motivation of Project

The aim of an online shopping system is to create a user-friendly platform that facilitates buying and selling goods over the internet. This involves developing a virtual storefront where customers can browse a wide range of products, view detailed descriptions, and conveniently search for specific items. The system should also provide secure payment gateways and efficient order management for a smooth transaction process. Ultimately, an online shopping system aims to benefit both consumers by offering convenience and competitive prices, and businesses by expanding their reach and increasing sales.

1.2 Project Objectives

The main purpose of this request is to build an online platform for operating our business processes. These selling activities were operated using a social media page, which is Facebook page. With this newly developed online platform, we would make more profits by operating, buying and selling transactions easily, efficiently, and conveniently. In the previous system, the following **problems** were found:

1. Customers often find it difficult and exhaustive to browse and look for the products that we sell, which requires scrolling through the entire Facebook page.
2. The seller has to answer the questions whether a specific product is in stock or not, that is, the availability of the products.
3. We need to ask the information of the customers every time they buy our products so that the delivery service can use it for transportation.

4. Some of the customer details are not static, changing from time to time.
5. We need to answer the questions which prompt basic information about the products to every customer, which are just displayed on the Facebook posts.
6. We need a separate effort to handle and track the orders.

Regarding the problems mentioned previously, we want a platform that possess the following **capabilities**:

1. The customers can easily browse and search for the products they desire.
2. The products are grouped into related categories by their hashtags.
3. The availability of each product is displayed easily visible to customers.
4. Customer information is collected and can be updated at any time on the system.
5. The order, delivery and payment tracking transactions are handled efficiently and effectively.

Above all business needs and requirements, the following **constraints** must be placed while the system is being constructed.

1. **Payment –** the payment can be made by either using mobile payment apps, cards, or cash on delivery. But the payment record must be stored for future reference.
2. **Delivery** – using other payment methods when receiving the cash-on-delivery products are not allowed. Canceling the orders after the delivery of orders with cash-on-delivery is not permitted.
3. **System –** the overall system must work with mobile payment services, bank systems for the usability of the customers while making the payment.
4. **Products –** must fall under one of many categories.

1.3 System Description

The Treasurer's Hub Online Shopping System will be constructed as a website using Java Enterprise Edition (J2EE) framework. Furthermore, the system will exhibit the layered architecture of Model-View-Controller to get a rigid system structure.

1.4 Advantages of Project

Building this system, we, the sellers, can achieve high incomes by receiving more venues from customers. We would also be able to manage and track orders and product delivery efficiently and effortlessly. For the delivery service providers, they will easily process the orders and deliver the products to the customers.

Chapter 2
Static Diagrams

The software design model for Treasurer’s Hub OSS is generally divided into static diagrams and Dynamic ones. Static diagrams include the ones which represent the structural organization of the system, such as class diagram and object diagram. This chapter focuses on the discussion of those static diagrams in a detailed manner.

2.1 Class Diagram

At the heart of any online shopping model lies a complex network of interacting components. To visualize this structure and understand how these elements work together, class diagrams can be used. These diagrams, built using the Unified Modeling Language (UML), provide a blueprint for the system by depicting the various classes (think as building blocks) and relationships. Each class represents a core entity, such as a customer, product, admin , along with its attributes (data points) and methods (functions). By analyzing these interactions within the class diagram, this can give valuable insight into the flow of information, processing logic, and overall functionality of the online shopping model.

There are several steps to follow to achieve the complete class model. The Treasurer’s Hub OSS is first analyzed from the perspective of entities and relationships between them in the ‘conceptual model’. The entities in the conceptual model become classes in the ‘domain model’ and contain the discussion of the detailed attributes and associations. To achieve the complete class model, the scenarios in the use case model are deeply analyzed using sequence diagrams. The overall operational behaviors of classes from the sequence diagram are then incrementally added to the class model as ‘operations.’

Conceptual Modeling

As mentioned previously, the conceptual model emphasizes the entities involved in the system and the relationships amongst them. It initiates the work of class modeling by identifying which ‘things’ need to be considered in the system.

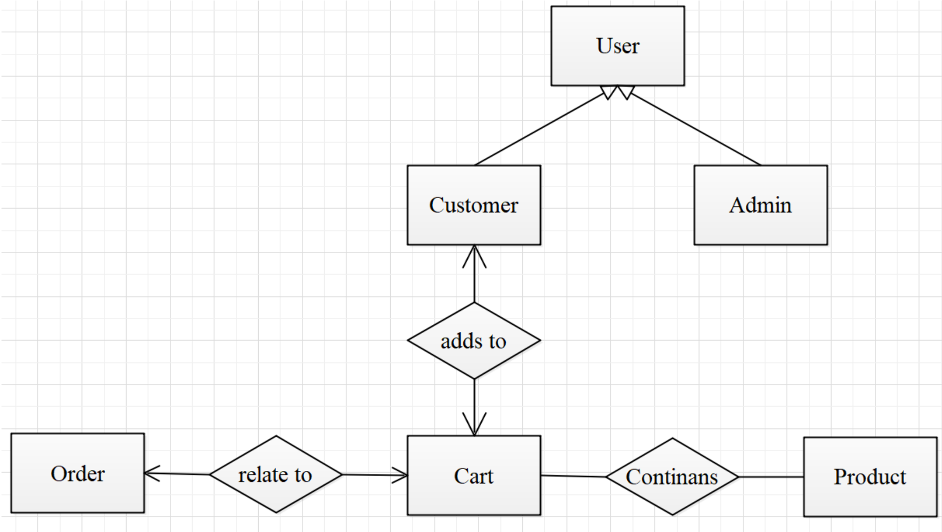


Figure 2.1 The Conceptual Model of Treasurer's Hub OSS

In **Figure 2.1**, the conceptual model in the form of Chen E-R notation can be seen. The fact that only the name of entity sets and relationship sets are described can be noticed. The detailed attributes in each entity will further be analyzed in the domain model.

Domain Modeling

The domain model is also called the preliminary version of class model. It only contains the classes, their attributes and associations amongst them. The relationship sets described in the conceptual model are represented as simple directed associations.

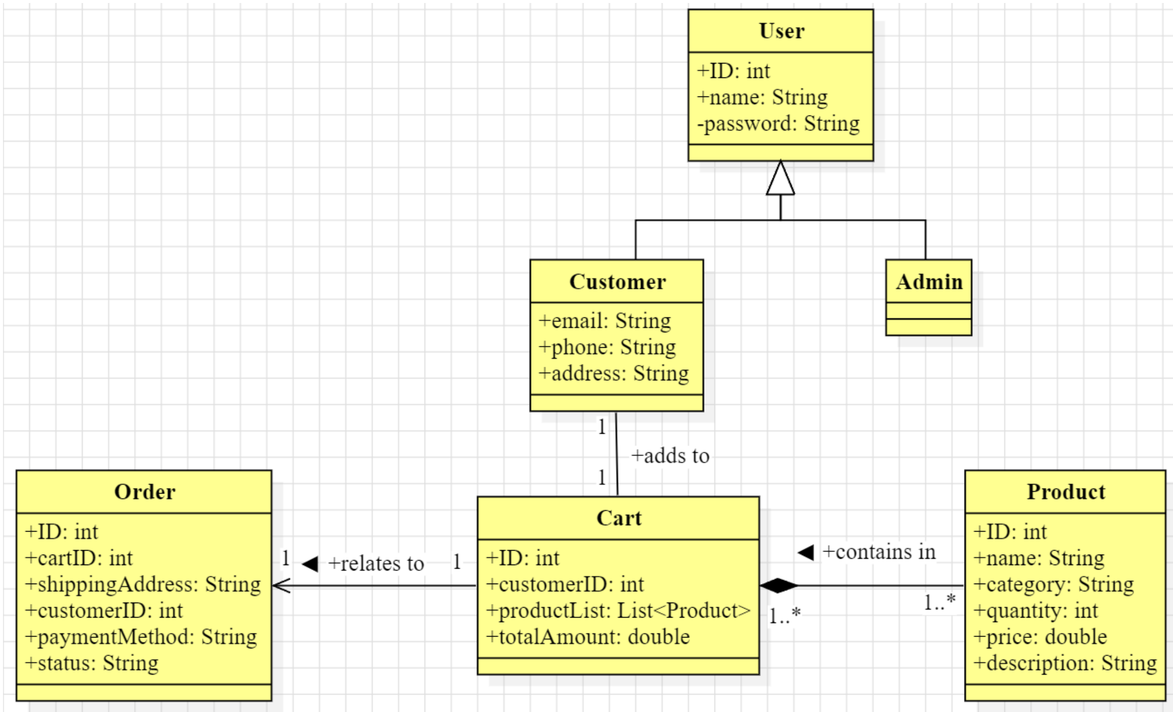


Figure 2.2 Domain Model for Treasurer's Hub OSS

Complete Class Model

The class diagram is incrementally modified by adding respective operations which are gathered from the behavioral model and in parallel with the use case diagram. As this is the software design model, the detailed data types of attributes, the return types and the constructors for classes are described. Since the system is intended to be implemented using the MVC architecture pattern using pure Java Enterprise Edition (J2EE), classes for the control and persistence layers become necessities. These classes are described in **Figure 2.3.**

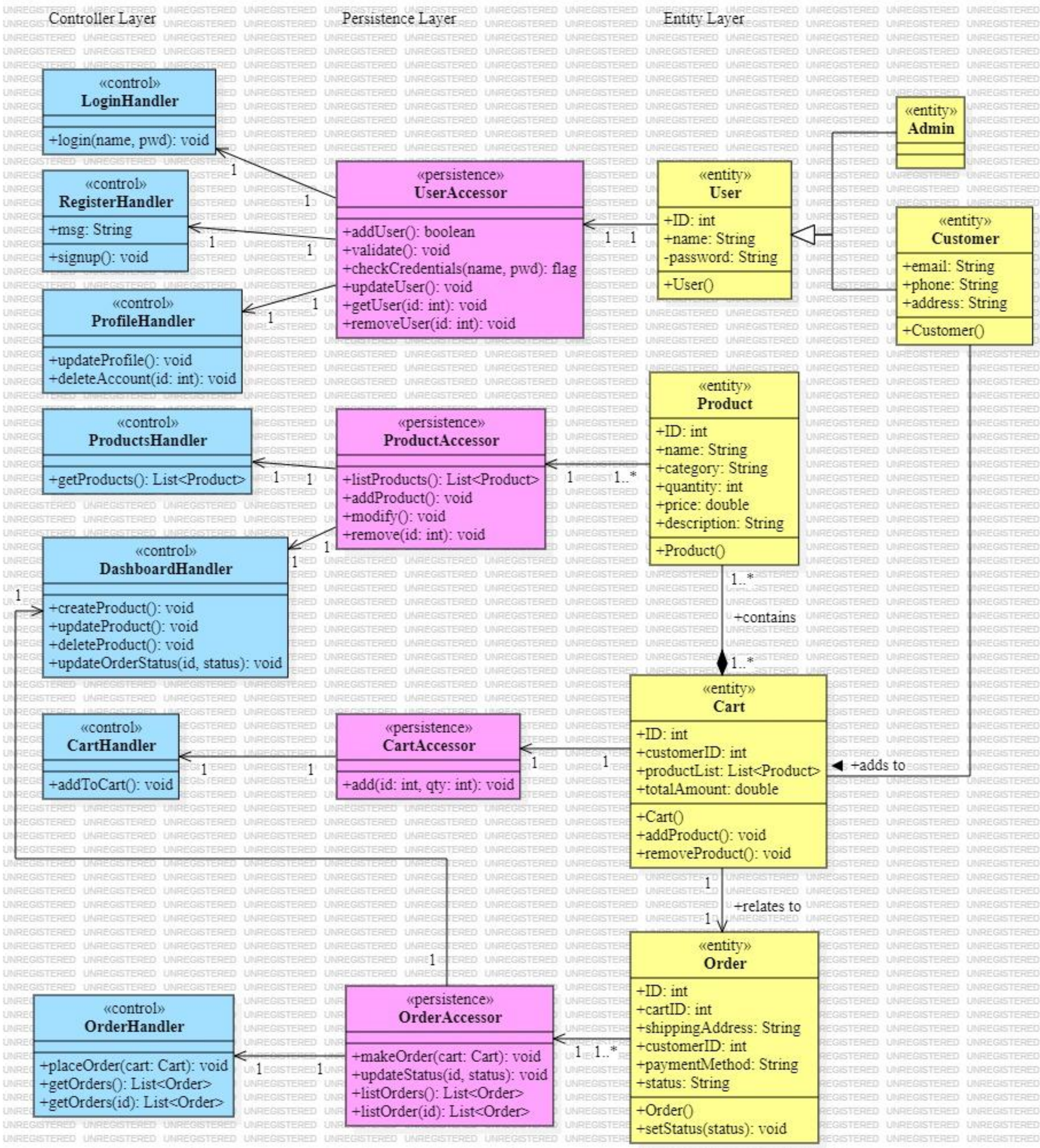


Figure 2.3 Complete Class Diagram for Treasurer's Hub OSS

2.2 Object Diagram

This section contains the description of possible objects which can be created from the design classes mentioned in Section 2.1 as can be seen in **Figure 2.4**. The links between the objects are just the instances of the associations amongst the classes in the design class model.

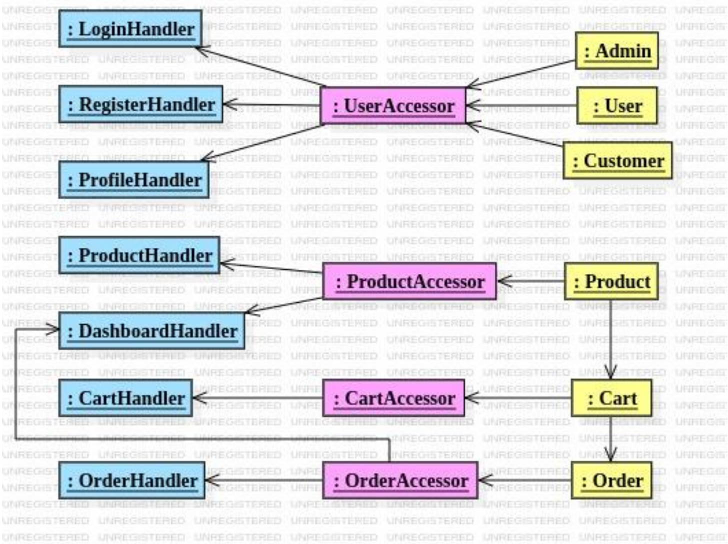


Figure 2.4 Object Diagram for Treasurer's Hub OSS

Chapter 3
Dynamic Diagrams

Having developed the static models, deriving the dynamic ones are now ready to be initiated. This chapter discusses parts of the design model, including the use case diagram, sequence and collaboration diagrams, and finally, statechart diagrams. These diagrams represent the behavior of the system elements and also known as the behavioral model.

3.1 Use Case Diagram

Treasures' Hub Online shopping model has revolutionized the way to purchase goods. To ensure a smooth and successful user experience, it's crucial to define the functionalities and interactions within this platform. Here's where use cases come in – they act as a blueprint, outlining the key actions and processes from a user's perspective. By understanding these use cases, this can help to create an online shopping model that caters to customer needs, fosters a positive experience, and ultimately drives sales.

As the first step to derive the use case diagram, the use case analysis is carried out to know what actors and use cases are needed. The result of use case analysis can be seen as follows.

Actor: Admin

User Profile:

- Login
- Manage Products
- Update Order Status

Actor: Customer

User Profile:

- Register
- Login
- Update Customer Info
- Purchase
- Add to Cart

The above user profiles only have the brief and preliminary overview of (base) use cases. The detailed analysis on the use cases shall be done using the UML use case diagram, followed by detailed descriptions for each use cases.

Use Case Diagram

Using the relationship between actors and user profiles of each, the Treasurer's Hub OSS can be represented by a UML use case diagram as in **Figure 3.1**.

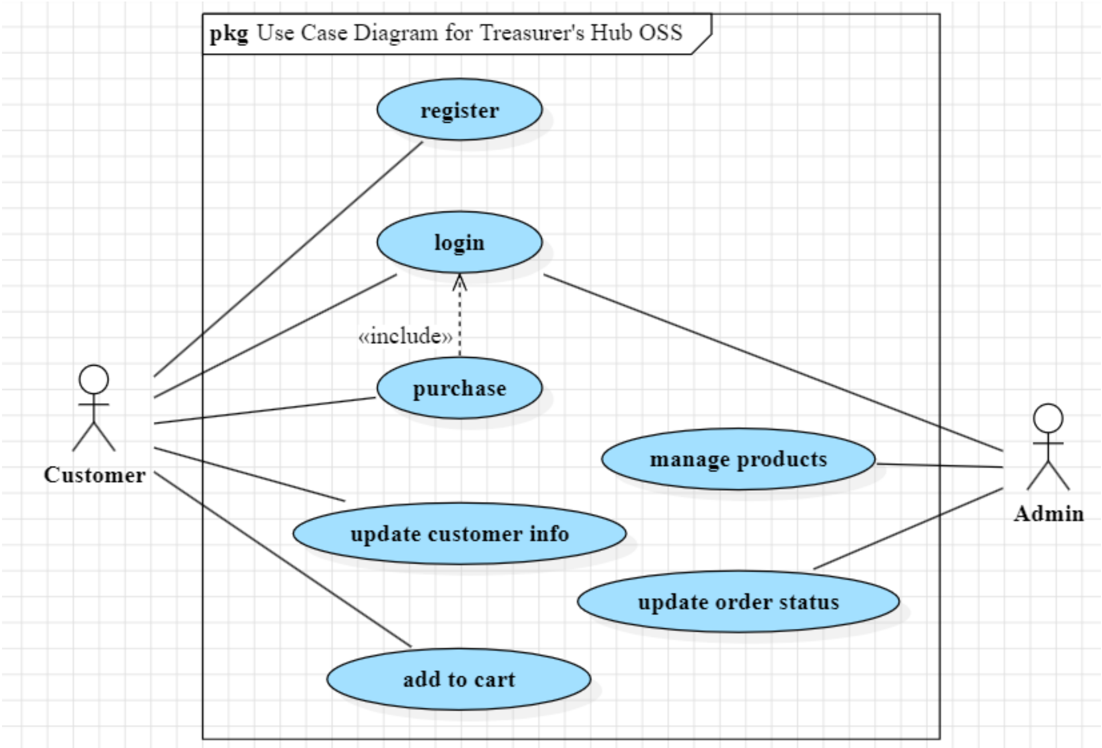


Figure 3.1 Use Case Diagram for Treasurer's Hub OSS

Use Case Descriptions

1. Register

- The customer navigates the Registration Page.
- The customer enters the necessary information.
- The customer clicks the "Register" button.
- The system directs the customer to the homepage.

2. Login

- The customer or admin navigates to the Login Page.
- The user fills the credentials in and clicks the 'login' button.
- The system validates the credentials.
- If the credentials are valid, the system navigates to the Admin Dashboard Page if the user is an admin, to the Customer Homepage if the user is customer.

3. Add to Cart

- The customer navigates to the Products Page or clicks on the desired category name on the side bar of the Customer Homepage.

- The customer selects the desired product item on the catalogue and sees the Product Info Pane.
- The customer clicks 'Add to cart' button, and that item is added to the shopping cart if the customer is logged in.
- If the customer is not logged in, the system navigates the customer to the Login Page.

4. Purchase

- The customer navigates to the Cart Page to see the items added.
- The customer checks the total amount and clicks 'Checkout' button.
- The system routes the customer to the Checkout Page.
- The customer is presented with the order information.
- The customer checks the shipping address and edits it in case it is different from the address registered.
- The customer chooses the preferred payment methods and makes payment.
- The system shows the success message once the payment is completed.

5. Update Customer Info

- The customer navigates to the Profile Page.
- The customer edits the desired information.
- Then click the "Update Info" button and the information is also changed on the system.

6. Manage Products

- Once logged in, the admin is navigated to the Admin Dashboard Page.
- The admin can enter necessary product information to the Add Products section and submits to <u>create new products</u>.
- The admin can <u>search or filter</u> the existing products in the Products section.
- To <u>update</u> the product information, the admin selects the desired product and edits new data on the 'Product Info pane,' and submits it.
- If the admin wants to <u>delete</u> a product, the selected product can be deleted by clicking the 'Delete' button on the Product Info pane.

7. Update Order Status

- Once logged in, the admin sees Order Lists section under the Admin Dashboard Page.
- The admin selects the desired order record and clicks the 'Change Status' button.
- The admin chooses from the desired status and the system automatically updates the order status.

3.2 Sequence Diagrams

The sequence diagram shows the details of the operation between the classes from there layers. It depicts how the operations are executed in each use case described in the use case diagram. This section contains the sequence diagrams in a complete manner.

Sequence1: Register

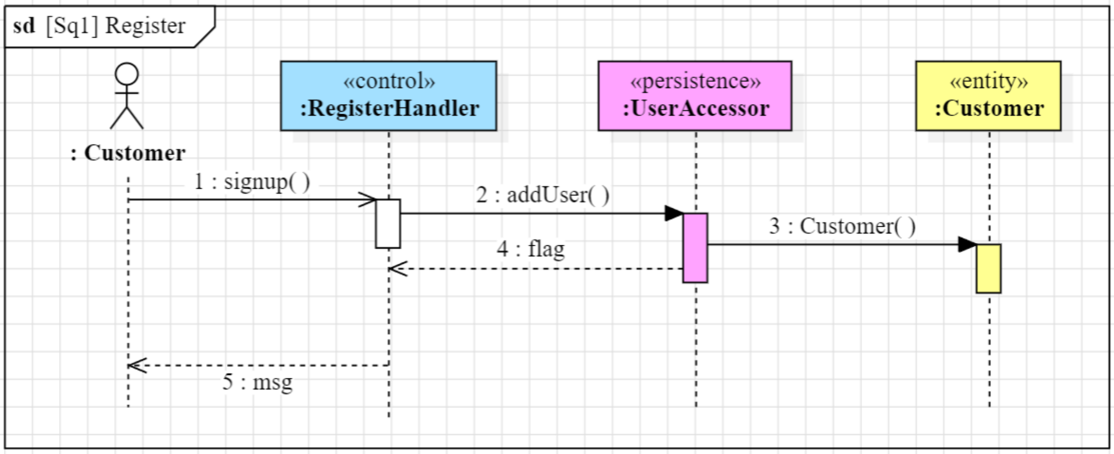


Figure 3.2 Sequence Diagram for Register

Sequence2: Login

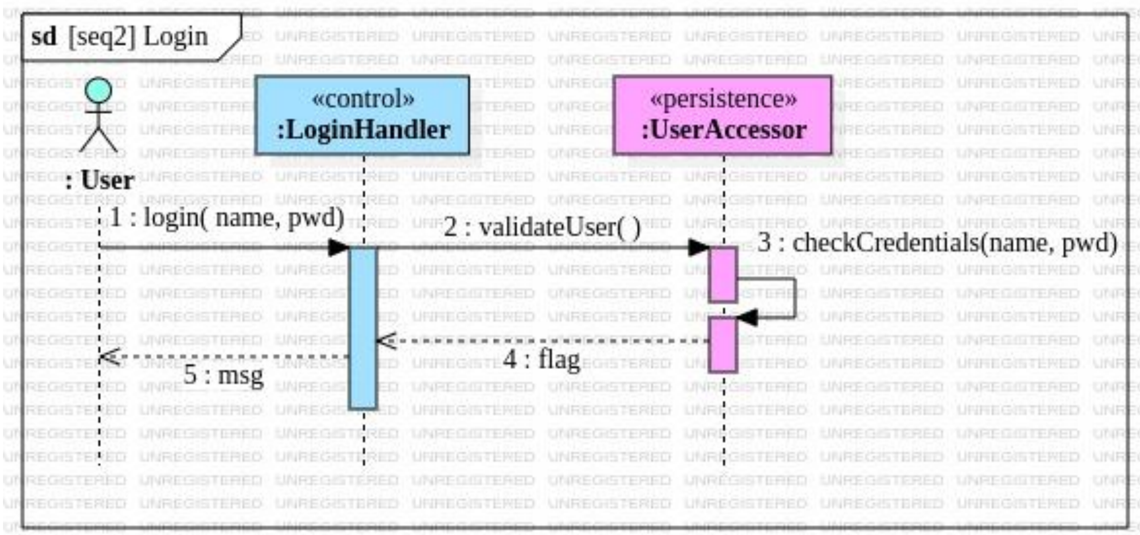


Figure 3.3 Sequence Diagram for Login

Sequence3: Purchase

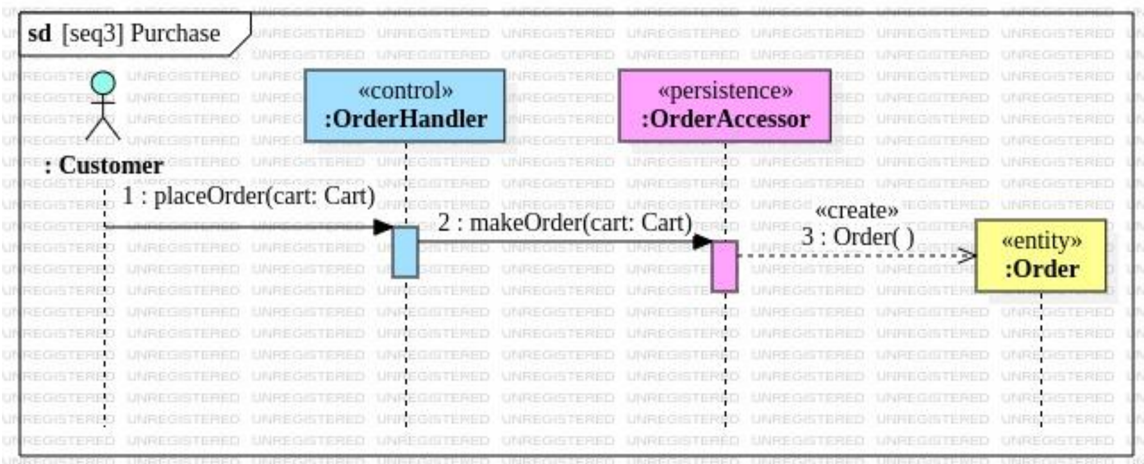


Figure 3.4 Sequence Diagram for Purchase

Sequence4: Update Customer Info

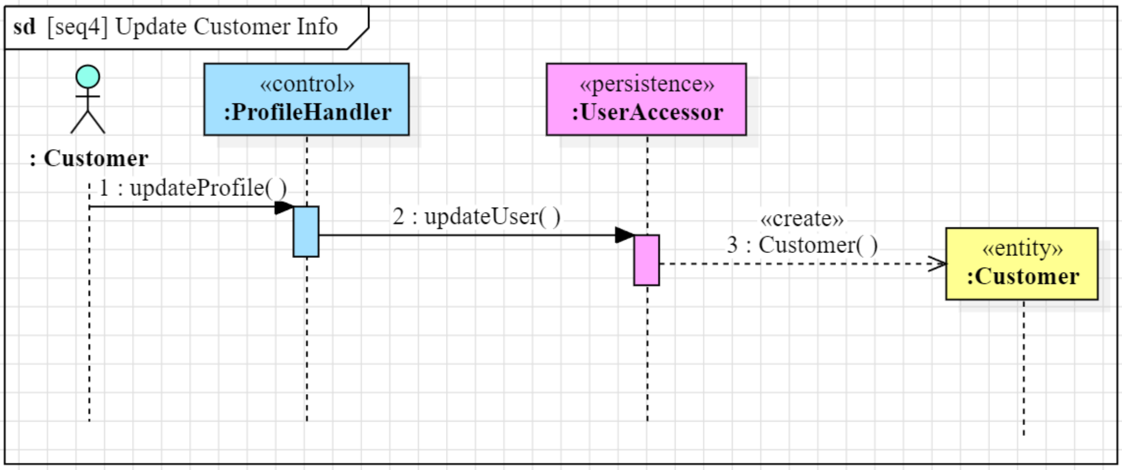


Figure 3.5Sequence Diagram for Update Customer Info

Sequence5: Add to Cart

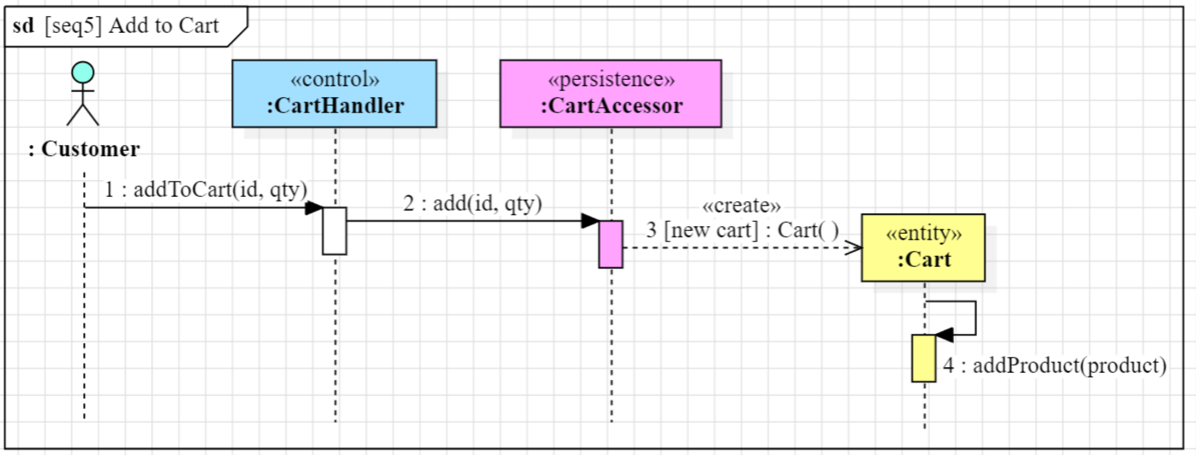


Figure 3.6Sequence Diagram for Add to Cart

Sequence6: Manage Products

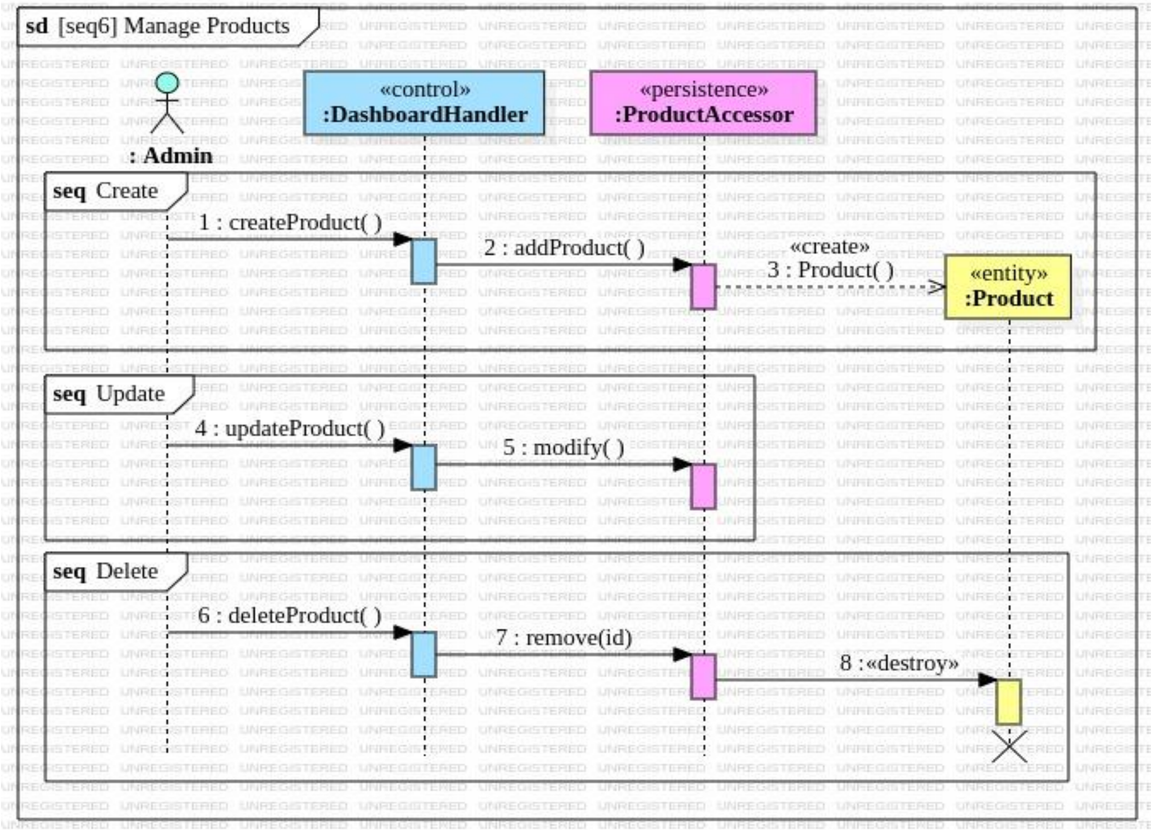


Figure 3.7Sequence Diagram for Manage Products

Sequence7: Update Order Status

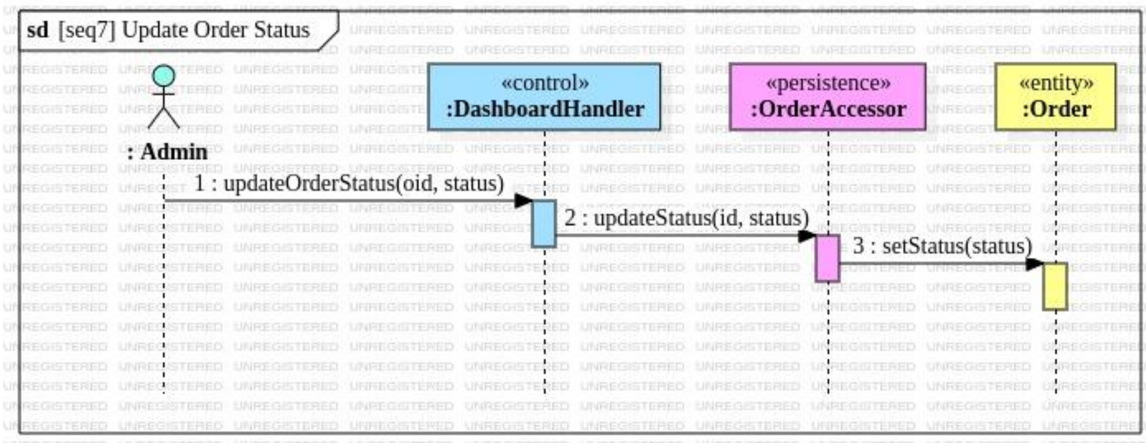


Figure 3.8 Sequence Diagram for Update Order Status

3.3 Collaboration Diagrams

Another type of interaction diagram which describes the system behavior is the collaboration diagram. It represents how the objects of classes in the design class model collaborate with one another to carry out the use cases. This section contains separate collaboration diagrams which are derived from the sequence diagrams.

Collaboration1: Register

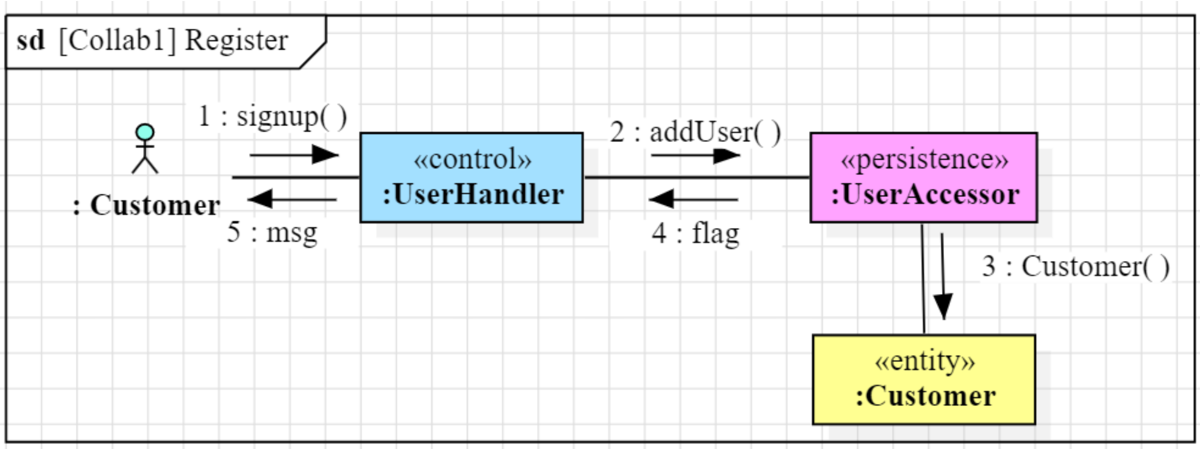


Figure 3.9 Collaboration Diagram for Register

Collaboration2: Login

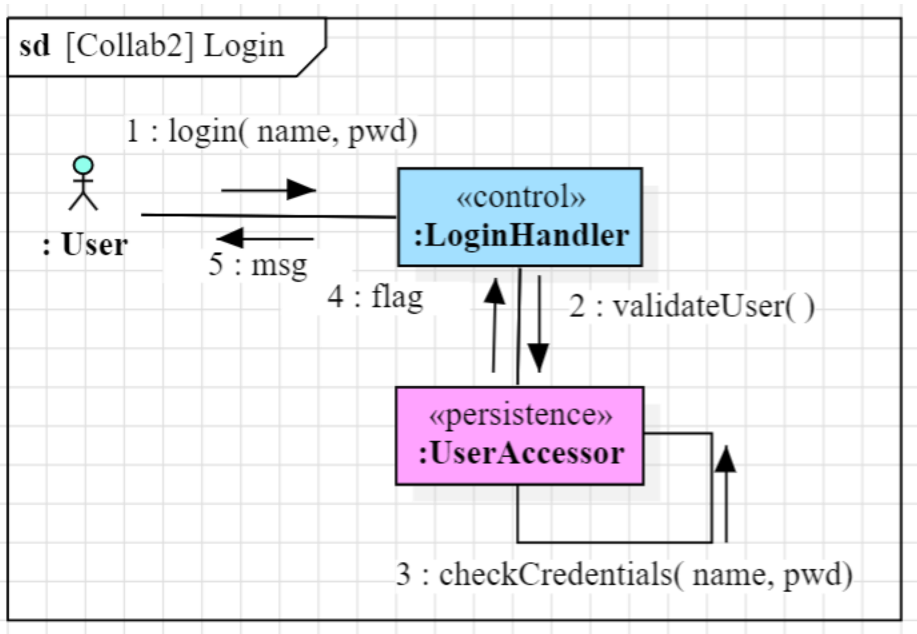


Figure 3.10 Collaboration Diagram for Login

Collaboration3: Purchase

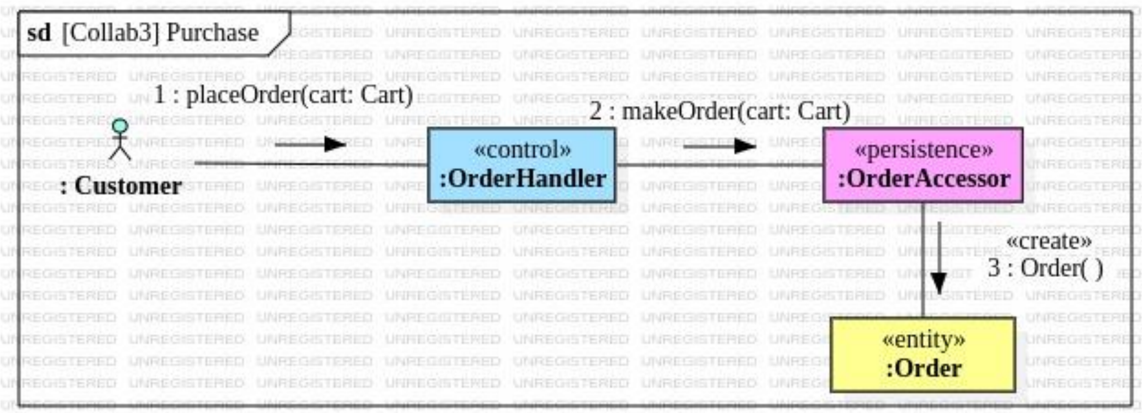


Figure 3.11 Collaboration Diagram for Purchase

Collaboration4: Update Customer Info

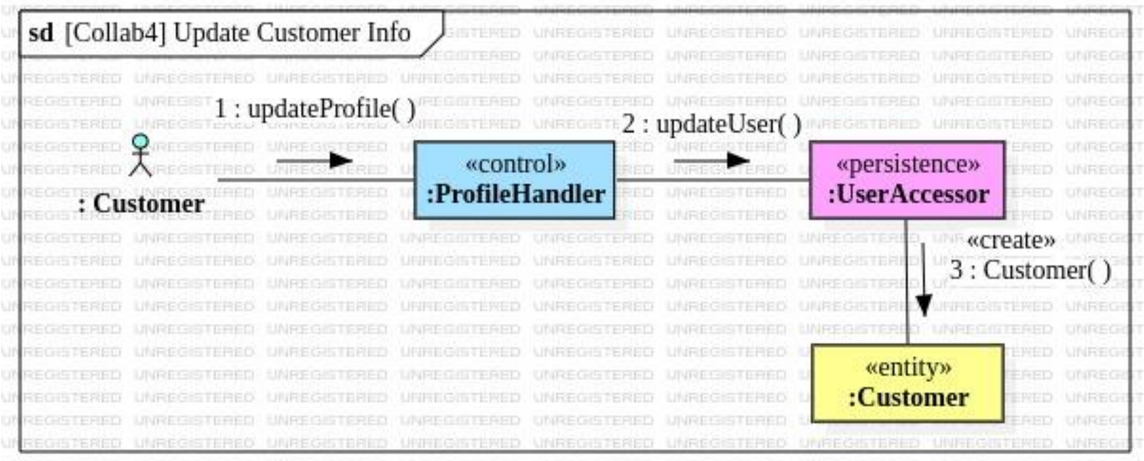


Figure 3.12 Collaboration Diagram for Update Customer Info

Collaboratioin5: Add To Cart

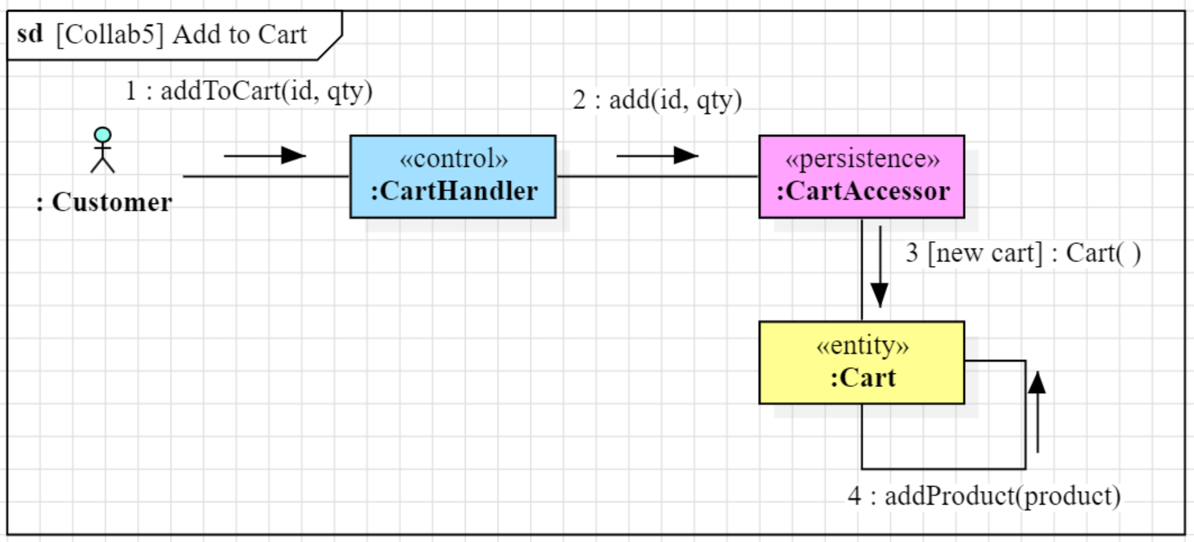


Figure 3.13 Collaboration Diagram for Add to Cart

Collaboration6.1: Create Product

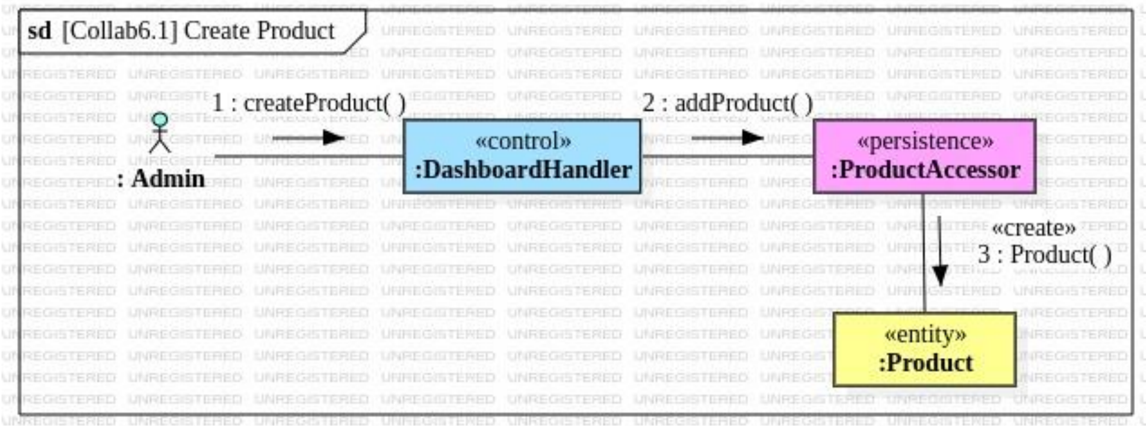


Figure 3.14 Collaboration Diagram for Create Product

Collaboration6.2: Update Product

Figure 3.15 Collaboration Diagram for Update Product

Collaboration6.3: Delete Product

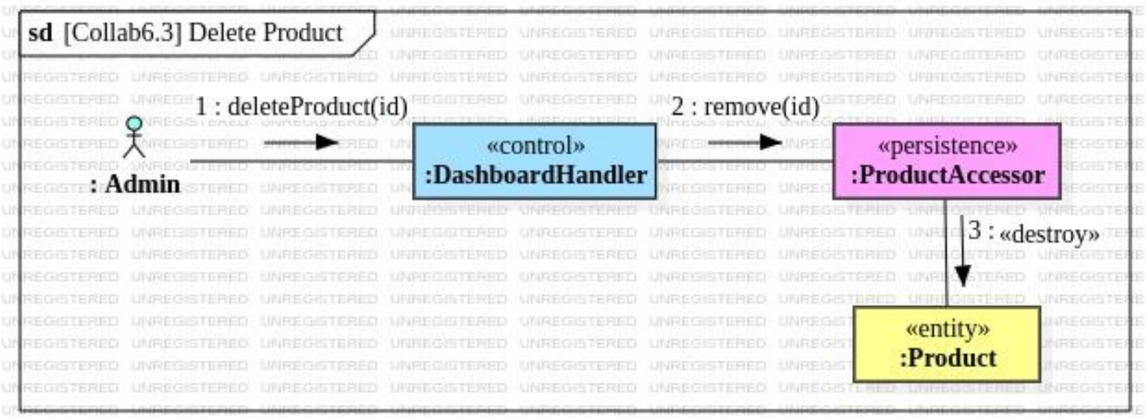


Figure 3.16 Collaboration Diagram for Delete Product

Collaboration7: Update Order Status

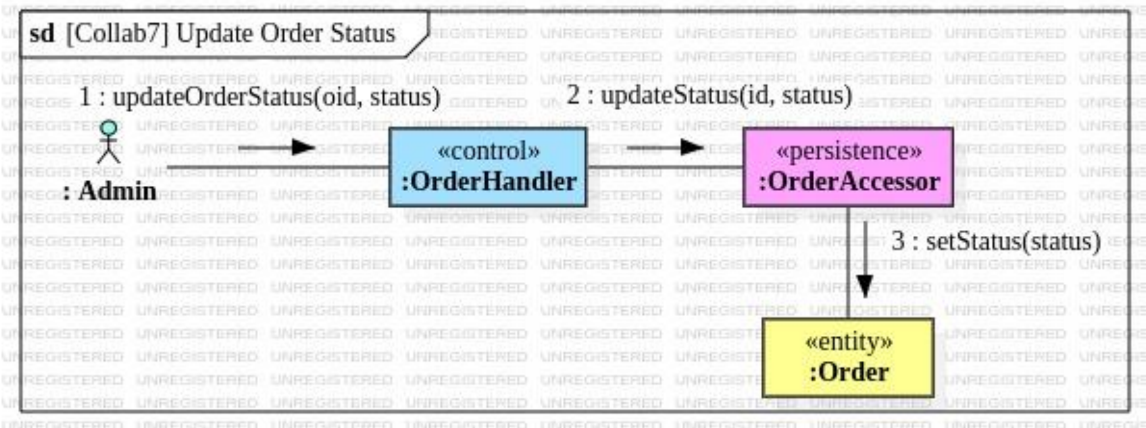


Figure 3.17 Collaboration Diagram for Update Order Status

3.4 Statechart Diagrams

Statechart diagrams have the ability to describe how objects of classes from design class model response to different events and what states they can exhibit. For the entity classes for the Treasurer's Hub OSS, possible states for each class are analyzed along with potential external or internal events they could receive, and how the objects respond to them. This section contains the results of that analysis work, which are a set of Statechart diagrams for entity classes.

Statechart1: Customer

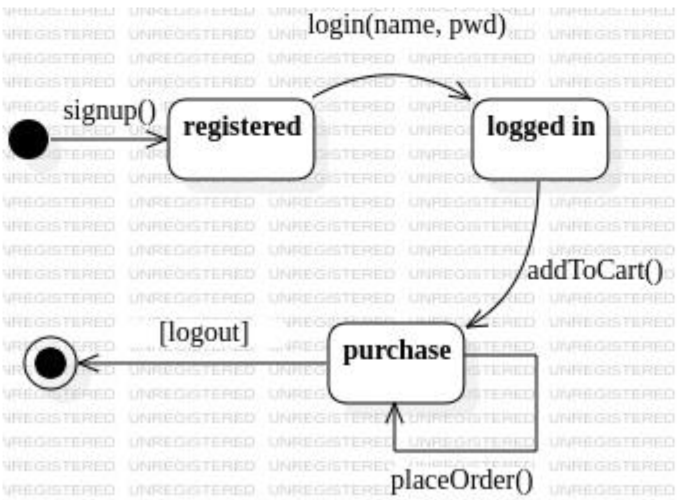


Figure 3.18 Statechart for Customer

Statechart2: Admin

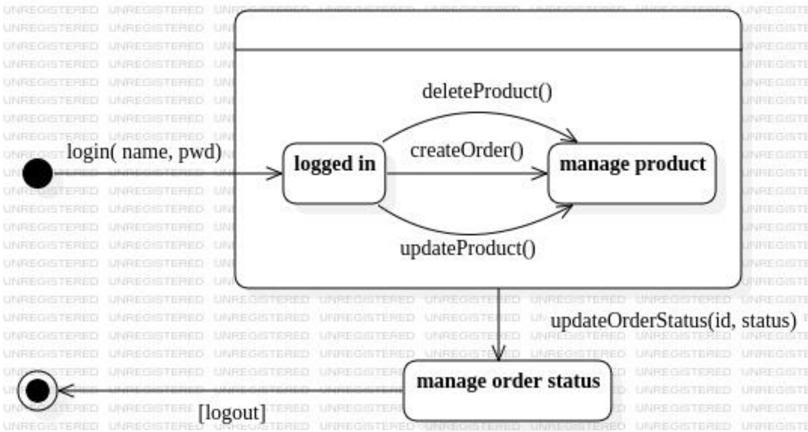


Figure 3.19 Statechart for Admin

Statechart3: Product

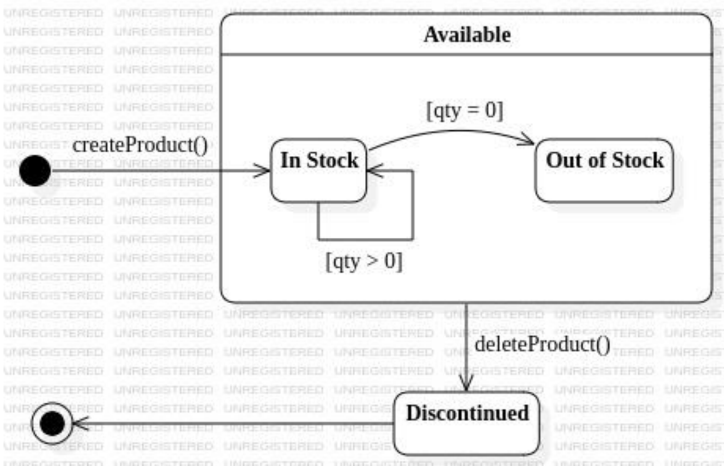


Figure 3.20 Statechart for Product

Statechart4: Cart

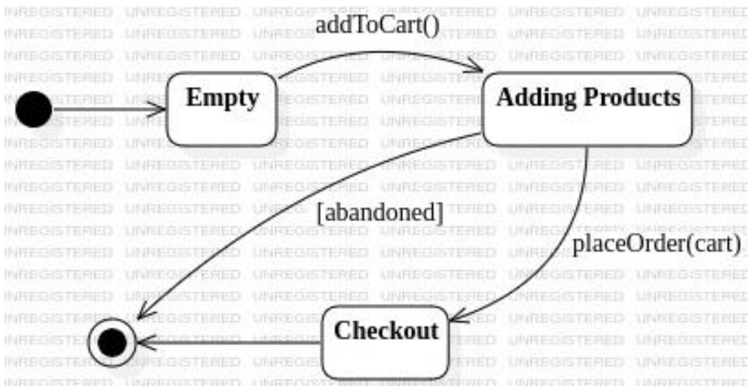


Figure 3.21 Statechart for Cart

Statechart5: Order

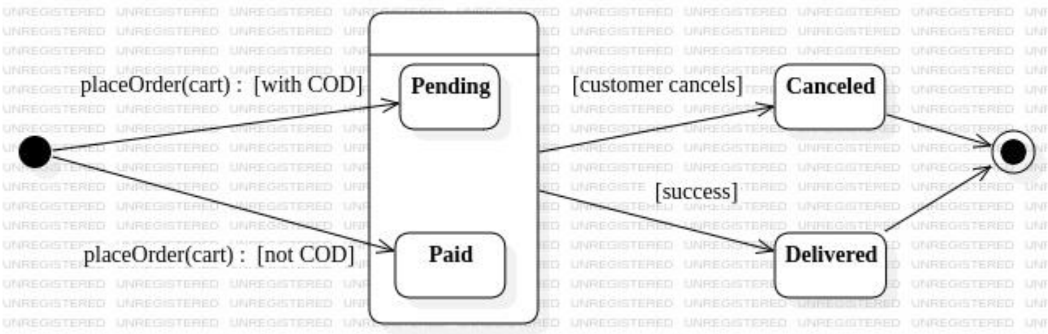


Figure 3.22 Statechart for Order

Chapter 4
Implementation Diagrams

After modeling the structural and behavioral aspects of the system, it is now ready to start constructing the implementation model. The implementation model describes how the software components would be deployed in the operating environment for the system. This chapter discusses the parts of the implementation model, which are component diagram and deployment diagrams.

4.1 Component Diagram

As its name suggests, the component diagram identifies all possible software components in the system which would work together to become a whole working system. Since Treasurer's Hub OSS is intended to be constructed with pure J2EE framework, the components are classified into separate folders, namely controllers, DAOs (short for Database Access Objects), utilities, and entities which can be seen in **Figure 4.1**.

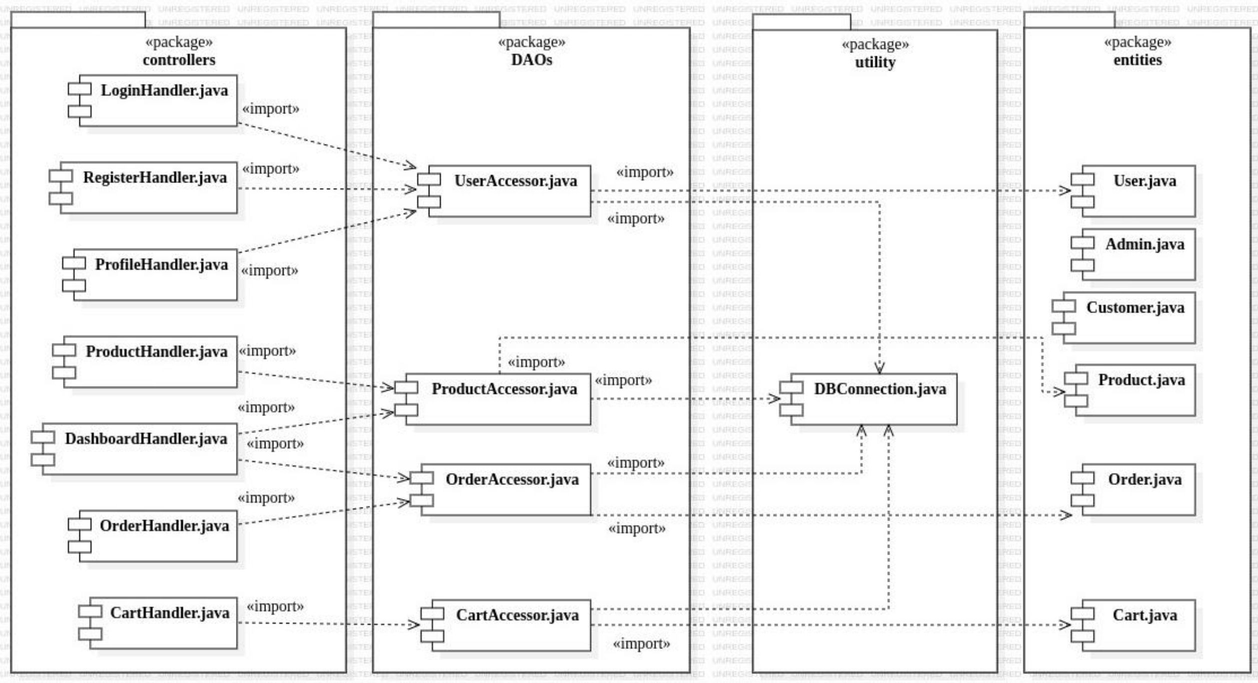


Figure 4.1 Component Diagram for Treasurer's Hub OSS

4.2 Deployment Diagram

The deployment diagram shows how the components mentioned are to be deployed in the operating environment. For Treasurer's Hub OSS, the Apache Tomcat Server is selected to be the one to deploy. Since this system is designed to run on a centralized server, all the components will be integrated on a single node as described in Figure 4.2.

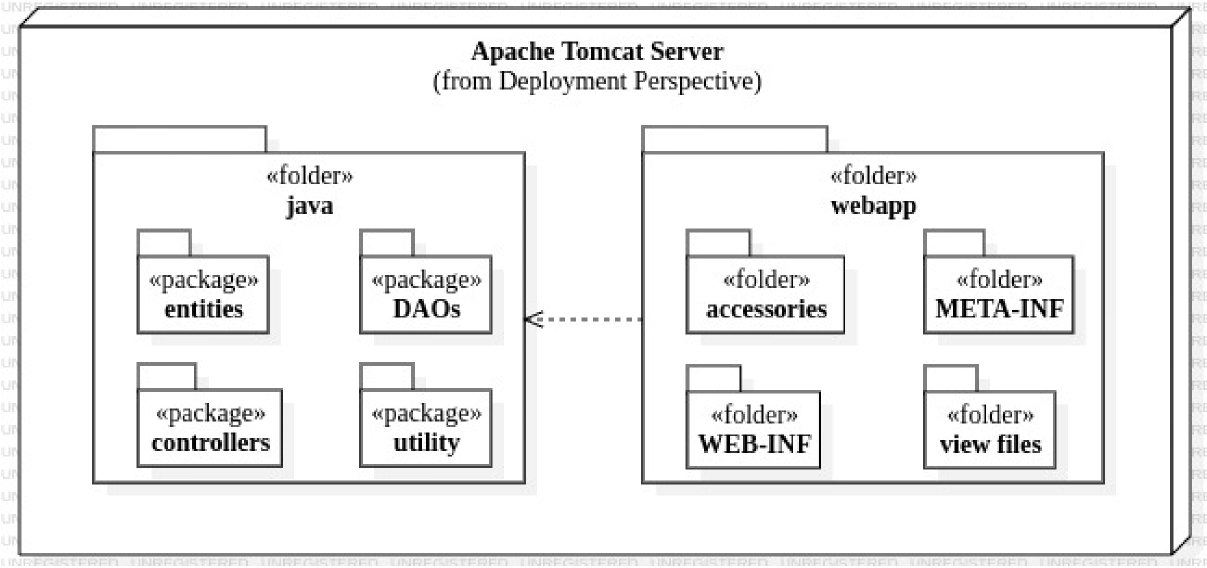


Figure 4.2 Deployment Diagram for Treasurer's Hub OSS

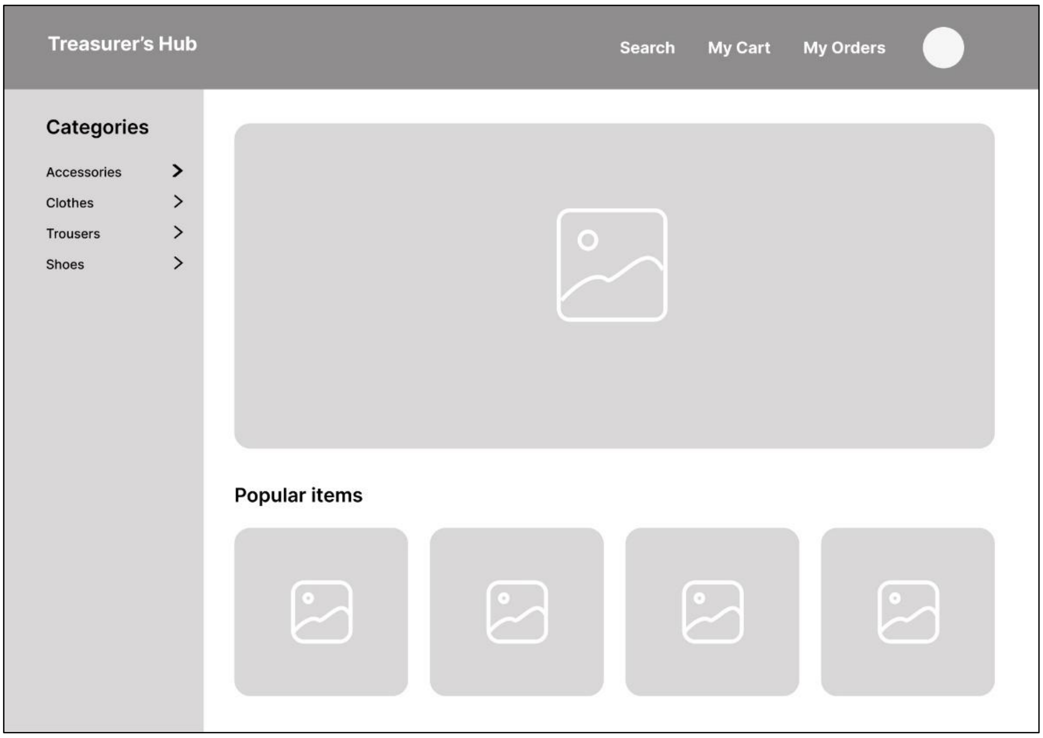
Chapter 5
Conclusion

Treasurer’s Hub Online Shopping System is started with the formation of a strong basic. It is initiated by acquiring the problem statement from the stakeholders, followed by the documentation of Software Requirements Specification (SRS). The SRS is then used as a foundation to start the requirements analysis modeling process. Now the analysis model derived from that is refined into a complete software design model which can be used as a blueprint for the actual implementation. As a future work, Treasurer’s Hub Online Shopping System is fully prepared to be coded using J2EE framework.

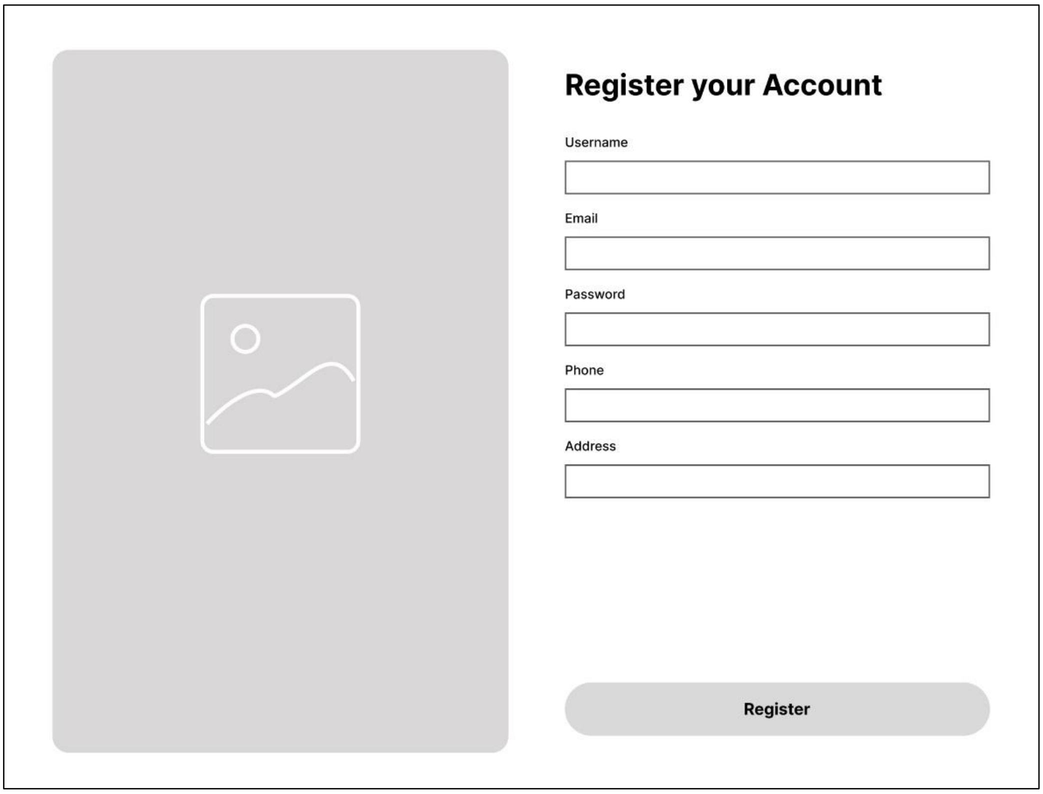
Appendix
Graphical User Interface for Treasurer's Hub OSS

This section is dedicated to the proposed graphical user interface for Treasurer's Hub OSS, comprising of the outputs from UI/UX modeling work using Figma.

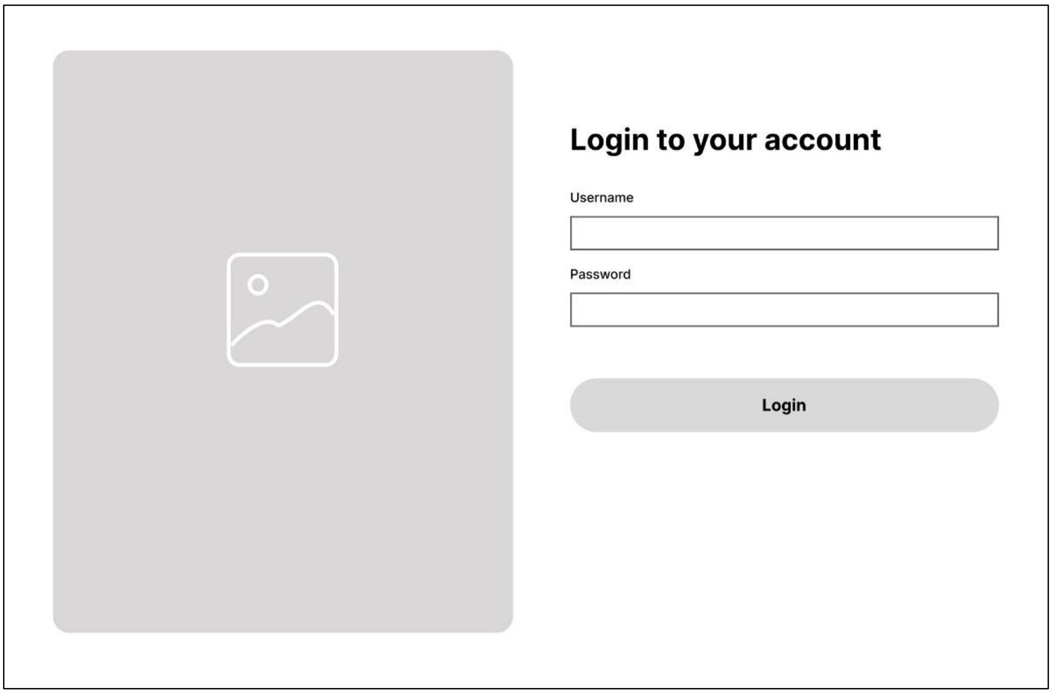
1. Customer Homepage



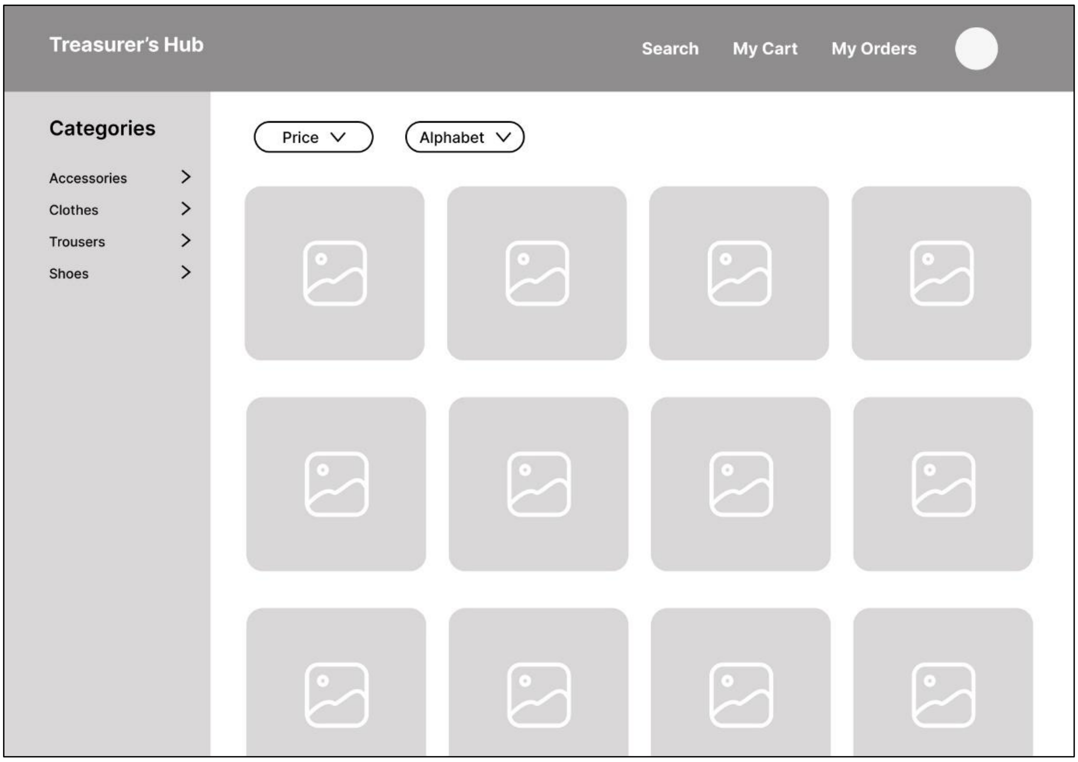
2. Register Page



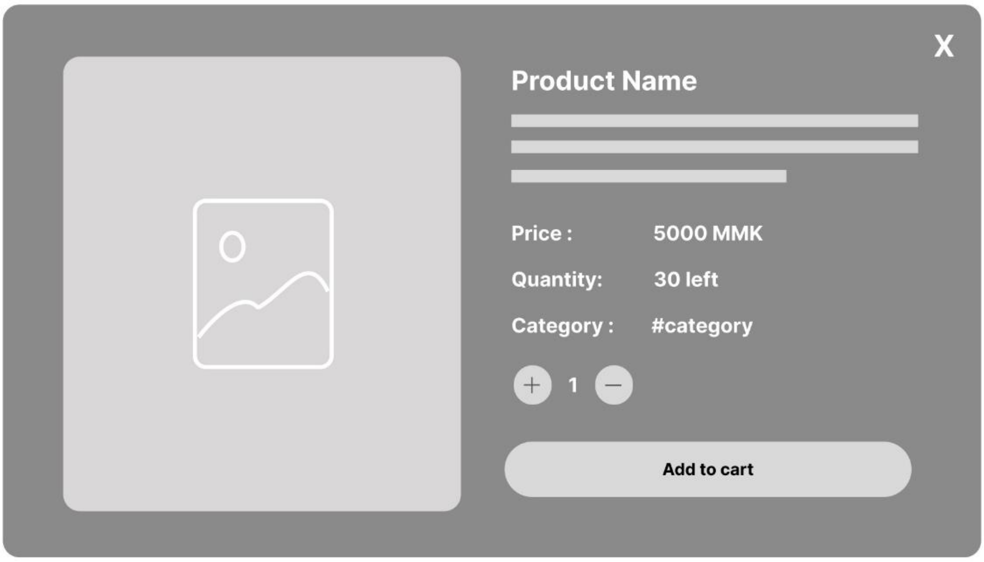
3. Login Page



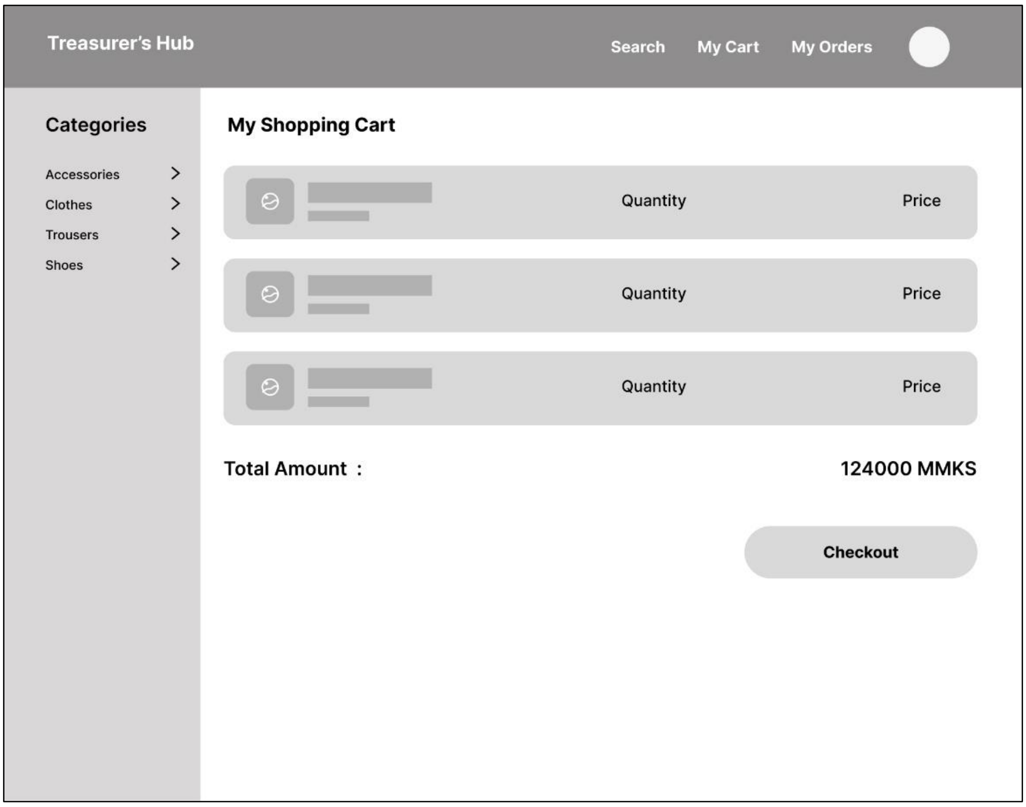
4. Products Page



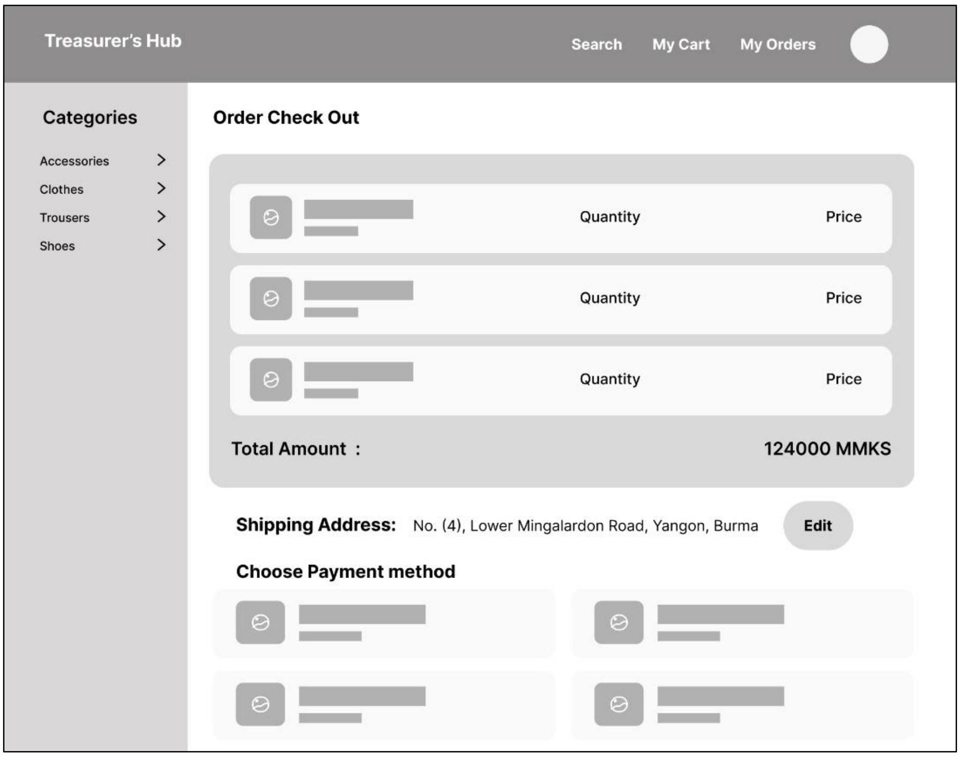
5. Product Info Pane



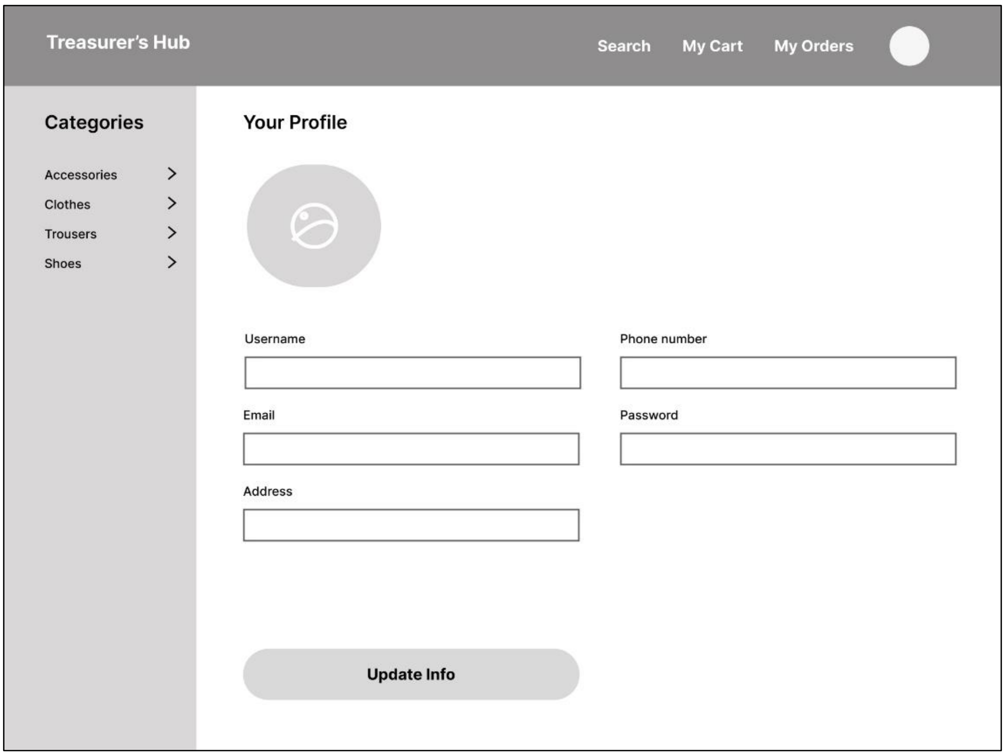
6. Cart Page



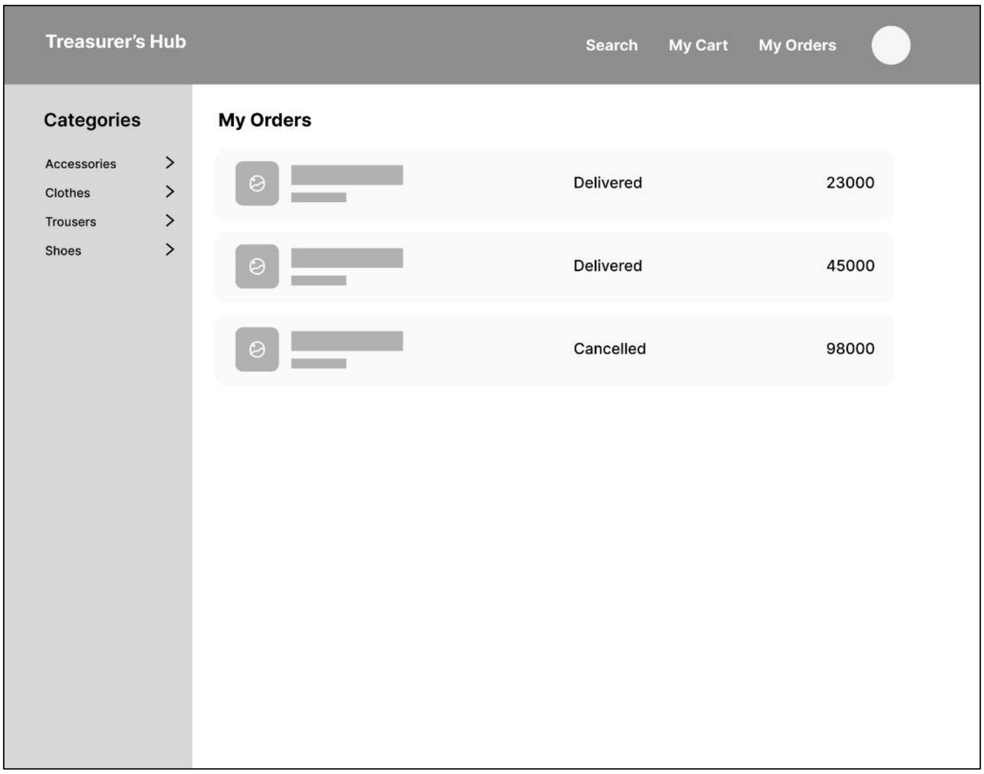
7. Checkout Page



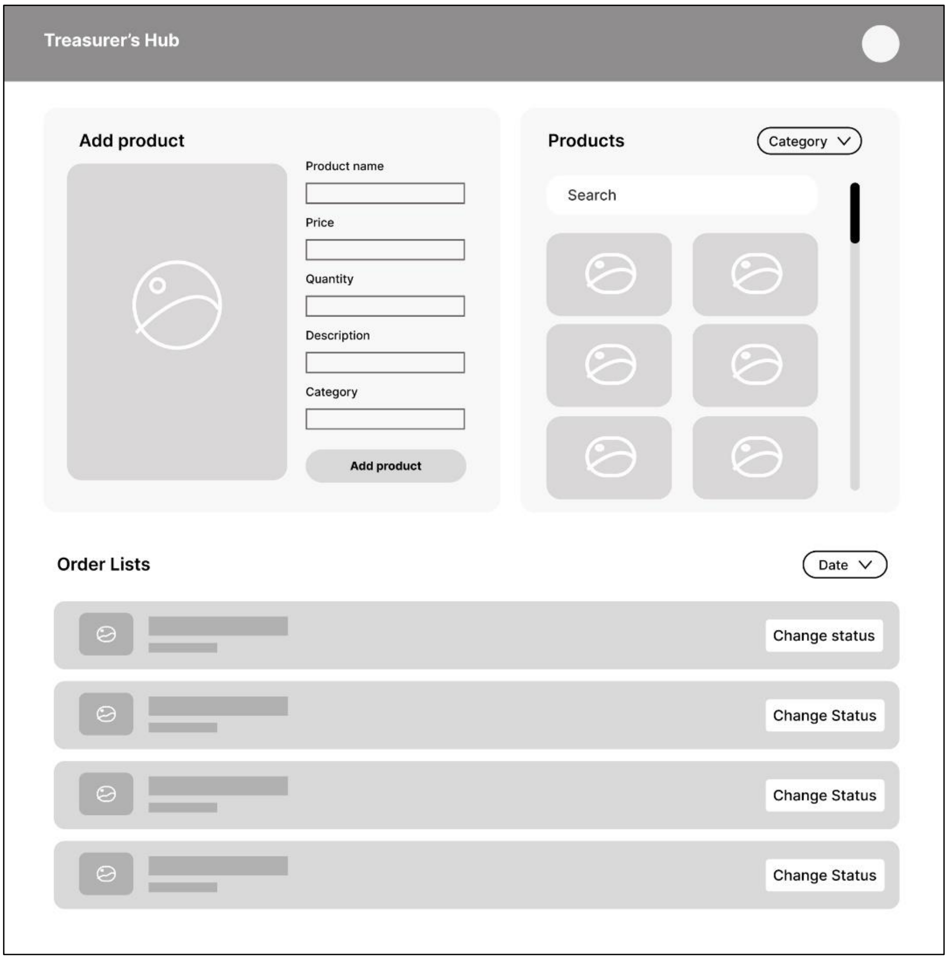
8. Profile Page



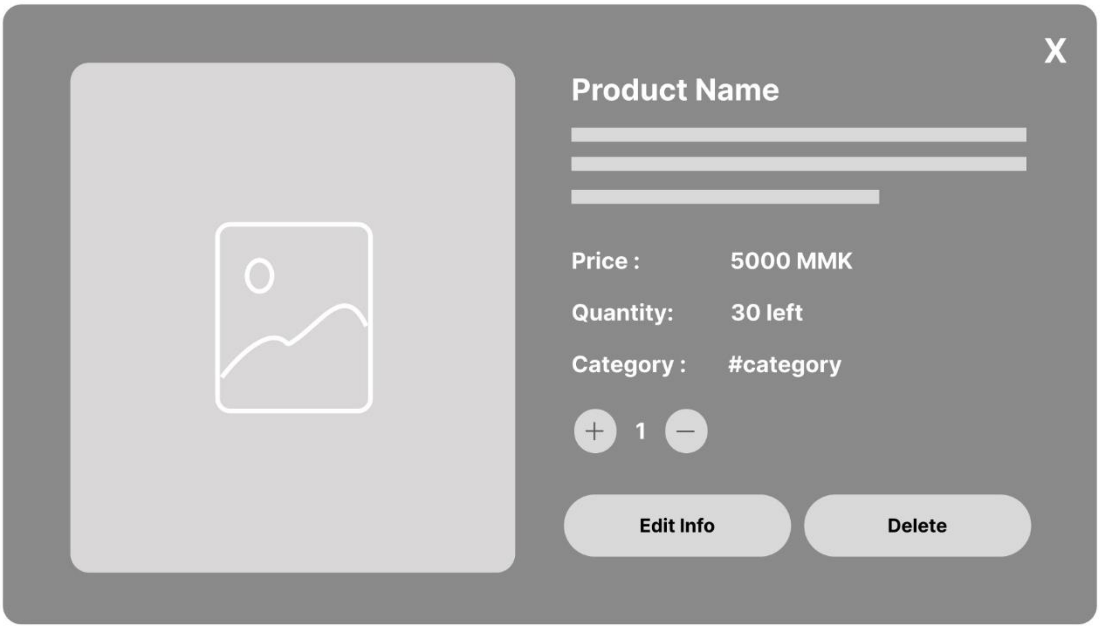
9. Orders Page



10. Admin Dashboard Page



11. Manage Product Info Pane



References

1. Pressman, R. S. (2020). *Software Engineering A Practioner's Approach* (9th ed.). Singapore: McGraw-Hill Education.

2. Priestly, M. (2004). *Practical Object-Oriented Design with UML* (2nd ed.). Singapore: McGraw-Hill Education.